

Programming Assignment on Programming with Threads

Implement a C program where the main thread creates five new working threads. Main thread will randomly select a nonnegative integer n in the range [10.000...50.000].

The program must compute $\sum_1^n i$. Each working thread will compute a partial sum, thus numbers 1 through n must be equally shared between working threads. Total sum will be calculated by adding partial sums, i.e., when each worker thread completes computing its partial sum it must add the partial sum computed to a global variable `sum`. Main thread waits for the worker threads to terminate and prints the sum.

By the above description it must be obvious that code that adds partial sums to global variable `sum` is a critical section that must be protected.

Your code must handle errors returned by system calls.